

Intention-retention phenomenon for online banking services in Pakistan: an application of technology acceptance model and expectation continuance theory

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Abstract

Purpose – This study aims to analyze the phenomenon of the intention-retention journey of customers for online banking services in Pakistan, using the technology acceptance model (TAM) and external factors, with a focus on the behavioral aspects of both Islamic and conventional banking users.

Design/methodology/approach – This study adopted a quantitative research approach to analyze the intention-retention journey of customers for online banking services in Pakistan. The hypothesized model was examined by the partial least-square structural equation modeling approach.

Findings – The results revealed that the benefits and perceived risks are significant, making online banking services a worthwhile adoption. Perceived usefulness, perceived ease of use and subjective norms are essential in predicting the intention to use online banking services among Islamic banking users in Pakistan. Moreover, behavioral traits showed that personality acts as a significant moderator, and planning is also an essential mediator between intention to use and actual usage. In addition, additive variables such as service quality act as a moderator, moderating the relationship between satisfaction and continuous usage. However, satisfaction and service quality are also significant variables, and they are sufficient to predict continuous usage directly.

Research limitations/implications – The most prominent characteristic of the current study is the behavioral aspects of customers, as observed in the pre- and post-analysis. The analyzed model in this study highlighted the antecedents of driving or constraining online banking services, as well as the process of moving from intention to use to continuous usage. This planned model extended the literature on the theories related to technological adoption. This study will help managers measure the external and behavioral traits, as well as customers' ability to use e-services.

Originality/value – A gap exists in research streams regarding the application of TAM in conjunction with other behavioral theories. Furthermore, the aforementioned technological acceptance model of the broadband life cycle has not been developed in any single study yet. Therefore, to fill the gaps by synthesizing previous studies, this study has been constructed to analyze the intention-retention cycle of customers.

Keywords Intention-retention journey, Augmented TAM, Online banking services, Technology adoption theories

Paper type Research paper



1. Introduction

In the global era of technology transformation in the banking industry, the introduction of e-services, organized infrastructure and digital innovation in financial markets has led to a mutiny of virtual transactions (Chatterjee and Bolar, 2019; Al-Amin *et al.*, 2023). Because technological developments empower the banking sector to acquire more advancements and convenience in their banking operations, especially for customers. Globally, the number of online banking users is projected to reach 3.6 billion (Source: www.statista.com). Even though over 85% of banking customers prefer using digital banking channels to conduct their financial transactions (Source: www.mckinsey.com). Despite the perceived usefulness (PU) of online banking services, customers are still reluctant to use them. The most significant concerns of customers when using online banking services are security and the theft of their personal information (Source: www.mckinsey.com). Consequently, risk is an imperative element involved in the practice of online banking which has a direct influence on the intention to use any services (Arif *et al.*, 2016; Afshan *et al.*, 2018) underlined specific constructs of risk in the framework of Internet banking users while there are further dimensions of risks that should have been highlighted to conclude the risk perception of users of online banking. Supporting this, Tarhini *et al.* (2017) and Putra *et al.* (2022) revealed a research gap in the concept of “perceived value,” which denotes the comparison of perceived benefits over perceived risks, leading to the adoption and continued use of digital banking. Literature reviews identified a gap in the direct impact of perceived benefits on behavioral intention, as most studies are limited to trust and risk factor models, similar to those of Arif *et al.* (2016) and Shahid *et al.* (2018). In addition, the rapid growth in Islamic banking, with the emergence of e-banking services, presents an opportunity to identify behavioral determinants for the intention to adopt online banking services. Furthermore, the acute shortage in the literature on behavioral traits of online customers of Islamic and conventional banks, such as studies of King and He (2006), Lee and Lehto, (2013), Rondán-Cataluña *et al.* (2015), Balochi *et al.* (2010), Aziz *et al.* (2016) and Sultan and Siddiqui (2010), focused on the performance of Islamic and conventional banking rather than customers’ behavioral and psychological perspectives. Therefore, a thorough analysis of the social construct of subjective norms is appealed to address the research gap.

As mentioned above, Al-Amin *et al.* (2023) highlighted several behavioral and psychological factors that influence the intention, adoption and retention of customers for virtual services. The deviation between intention and actual behavior is a natural phenomenon that exists due to the behavioral traits of any individual (Fishbein and Ajzen, 1975). These behavioral traits bridge the intention-behavior gap in decision-making and act as moderators and/or mediators. Sometimes, an individual with a strong intention to use technological services may not use them. In contrast, a person with a low intention becomes an adopter due to external factors that result from the intention-behavior gap phenomenon (Maier *et al.*, 2012). In the previous studies, Afshan *et al.* (2018), Pavlou (2003), King and He (2006) and Lee and Lehto, (2013), other researchers argued about the determinants of the behavioral intentions of users to predict the actual behavior of the consumer, while antecedents of actual behavior have not been highlighted yet in the literature. Previous frameworks (Fu *et al.*, 2018; Lestari *et al.*, 2024) have narrowly focused on two-dimensional risk perceptions in online banking. In contrast, the current study is acquainted with a more nuanced, granular and culturally contextualized risk dimension tailored to Pakistan’s volatile digital banking landscape.

Moreover, the literature is currently restricted to behavioral intention (Rondán-Cataluña *et al.*, 2015), while actual behavior, including moderator and mediation effects, has not been considered separately in any study. Integrating atop, the post-adoption phenomenon of any

product or service is another aspect of product development and acceptance. Most of the studies focused on pre-adoption behavioral aspects of using digital banking services, such as Laukkanen and Kiviniemi (2010), Zhou *et al.* (2021) and Al-Qudah *et al.* (2024). Even though there may be many factors affecting the decision to repurchase or reuse any product or service, literature revealed that service quality, satisfaction and experiences are highlighted in the previous studies to have an impact on the continuous usage of any product or service, such as Blut *et al.* (2015), Kitapci *et al.* (2014) and Chatzigeorgiou and Simeli (2017). In addition, Kaur *et al.* (2021) proclaimed satisfaction, service quality and convenience as mediators or moderators of customers' loyalty. Moreover, the significance of service quality is well established in the research field of online banking and is defined as the discrepancy between the expectations and experiences of online users. In the other dimension, service quality has been a dominant factor in imparting customer satisfaction and continuous usage (Lin and Wang, 2006). Considering the link between service quality and continuous adoption, some researchers have found an indirect effect of service quality on continuous usage (Yang and Jun, 2002). While, most of the previous studies engrossed only on service quality as an antecedent of satisfaction and pro element for customer retention equivalently to studies (Kassim and Abdullah, 2010; Donighi and Yousefi, 2015; Taylor and Baker, 1994; Saleem and Yaseen, 2017; Hussain, 2016; Namukasa, 2013). Moreover, Karimi *et al.* (2015) and Al-Amin *et al.* (2023) also discovered the need for pre-post purchase stages of customers' decision-making in e-services frontiers. Lăzăroiu *et al.* (2020) also determined in a study about limitations in the research momentum of social commerce platforms, and online purchasing intentions in the light of source credibility

Extracted from the above, a void remains in the literature that examines the connection between initial intention and long-term retention in the context of online banking services, despite the technology acceptance model (TAM) having been widely used to understand consumers' early intentions to adopt technology. At the same time, prior studies have integrated TAM with other behavioral theories to propose a multi-stage TAM-expectation-confirmation model (ECM) framework for the digital banking usage-continuous usage phenomenon, such as those by Fu *et al.* (2018), Rahi *et al.* (2021) and Nguyen and Ha (2022). Although, Nguyen and Ha (2022) extended TAM with an additional variable "trust" in their study to quantify intention to purchase online foods, whereas other behavioral constituents are neglected in the referred study. In addition, Cranney *et al.* (2025) and Marx *et al.* (2023) addressed a literature gap in their studies, highlighting how technological transformation enhances individual decision-making. This study addresses the highlighted research gap and introduces planning as a mediator, acting as a micro-mechanism that links the intention-retention journey of customers.

Moreover, literature streams reveal that the TAM application is restricted to behavioral intention. At the same time, actual behavior or usage, with moderator and mediation effects, have not been considered separately, especially in the banking industry. While, Rahi *et al.* (2021) exhibited a combo of TAM, expectation continuance theory (ECT), trust construct and self-determination theory, delineating the post-adoption phenomenon of online banking services. Furthermore, Khahar *et al.* (2025) presented the same model for cloud-computing services; however, neither study aligned the behavioral factors affecting usage behavior. Hence, this study serves as a showcase of a pro-version of the above studies, incorporating personality as a moderator and proxy for behavioral factors that determine the actual usage of online banking services.

Precisely, technological adaptation to its continuous use is a sequential process that arises from pre- and post-usage of technology. Furthermore, the aforementioned technological acceptance model of the broadband life cycle has not been developed in any single study to

date. Therefore, to fill the gaps by synthesizing previous studies, this study has been constructed to analyze the intention-retention cycle of online banking users in Pakistan.

2. Review of literature

A review of the literature encompasses both theoretical and empirical studies related to the proposed research. It commonly provides theoretical grounds, hypotheses development and an overview of previous studies for any research.

2.1 Theoretical literature

This section includes the theoretical background and rationale of the proposed modified TAM with the collaboration of other variables, hypothesized by the subsequent arguments as follows:

2.1.1 Technology acceptance model (TAM) leading to modified TAM. Precisely, the TAM acts as an archetypal model to predict the behavior of customers in accepting any technological advancement. [Davis \(1989\)](#) described determinants of intention to use any information technology that predict actual behavior toward the usage of online facilities. TAM assumes that an individual's behavioral intention to use any technological advancement can be measured by two main factors, perceived ease of use (PEOU) and PU. PU is defined as the "belief of a person that use of technology will augment his or her job performance or perform any specific task." At the same time, PEOU encompasses a degree of ease in operating prescribed technology. Several empirical tests in different studies have been conducted to determine the impact of these two factors, and PU is the strongest determinant of Intention to use any technology. In addition, PEOU is considered the second significant antecedent of Intention to use ([Venkatesh and Devis, 1996](#)). Subjective norms, or social influences, are considered an external variable, defined as a person's perception that people important to them think they should or should not perform a task ([Fishbein and Ajzen, 1975](#)), and are polled as a direct stimulus of intention to use technological advancements. The rationale for the direct effect of subjective norms on intention to use is that individuals initially feel hesitant about new technology but comply with it after being recommended by an important referent. Furthermore, in the Pakistani culture, where ideas are often adopted from peers, the eldest family member and other social or religious influencers, the significant impact of subjective norms and intention to adopt online banking may be explored ([Hussain et al., 2020](#)). In the same manner, perceived acceptance of important referent groups is the central reason in those contexts in which the Collectivism approach has been practiced. The rationale for including the subjective norm is to identify the diversity of cultural factors that influence technological transformation, especially in developing countries.

Perceived risks and benefits that affect consumers' decision-making to use any product or service. Many researchers have adopted the TAM in various contexts, sectors and cultures, incorporating different variables extracted from other theories to predict customers' behavioral intentions toward the adoption of technological advancements ([Shanab and Pearson, 2007](#)). The theory of risk elaborates on perceived risks as multidimensional, encompassing financial risk, performance risk, social risk, physical risk and privacy risk, which can vary according to the product and service. Theories and empirical studies have found that perceived risks are a significant element contributing to the intention to use online services ([Lee, 2009](#)). Parallel to this, online banking has been considered an effective and economical way to perform banking transactions. Perceived benefits, as another external variable, include lower transaction charges, higher deposit rates, discount offers, faster speed transactions and paper-free documentation. Co-jointly, monitoring authorized performance

and 24/7 are the most significant benefits of online services and are considered a preliminary factor of Technology acceptance.

From the perspective of e-banking, actual usage or behavior refers to the customer's self-reported use of technology services. Fishbein and Ajzen (1975) postulated that behavior does not always act according to intention, and some imperfections exist between intention and the actual behavior of a person due to external factors. These externalities act as mediators and/or moderators, causing the intention-behavior gap. The supporting theory of the intention-behavior gap (Robey, 1979) confirmed a strong relationship between personality traits, attitudes and beliefs and the deviation of the intention-behavior gap. Furthermore, planning involves a complex aspect of linking behavioral responses to specific circumstances and is categorized into action planning and coping planning. Action planning refers to actions related to specific criteria for performing the behavior (Sniehotta *et al.*, 2005), while coping planning covers the expectancies of barriers between intention and behavior. Hence, planning is hypothesized as a mediator between intention and actual behavior. Evidently, Reuter *et al.* (2008) and Norman and Aron (2003) conducted studies in which planning was observed and experimentally induced as a significant mediator between Intention and actual behavior.

Consequently, technological acceptance incorporates the repurchasing activity of any technological service or product. Continuous usage intention refers to the post-adoption behavior related to digital services, where the level of satisfaction can jointly determine it. The significance of service quality is well established in the research field of online banking and is defined as the discrepancy between the expectations and experiences of online users. In the other dimension, service quality has been a dominant factor in imparting customer satisfaction and continuous usage (Lin and Wang, 2006). Considering the link between service quality and continuous adoption, some researchers found an indirect effect of service quality on continuous usage (Caruana, 2002; Cronin and Taylor, 1992) while some proclaim a strong and direct impact; Yang *et al.*, 2004).

Ultimately, TAM, with its extensions such as TAM1 and TAM2, and the integration of TAM with other behavioral theories and constructs, has been widely applied in prior studies. Recently, Bhattacharjee and Lin (2015) and Chen *et al.* (2022) and Rahi *et al.* (2021) highlighted ECT with a combination of motivational and trust-centered factors to describe continuous usage. Connecting the research streams, Yang *et al.* (2023) marked that continuous usage in digital services does not depend on elementary constructs of TAM but is also associated with post-adoption satisfaction and inherent motivations. Besides, Nguyen and Ha (2022) and Cranney *et al.* (2025) found that tinted trust, which involves delegating data security and regulatory protections, is positively related to the ability to withstand usage. By synthesizing the above dimensions, the current model of this study bridges the intention-retention gap, building upon the critiques of TAM-centric studies that often focus on the post-adoption phenomenon.

2.1.2 Proposed collaborations in TAM. 2.1.2.1 Intention-behavior gap theory: planning mediates, personality moderates behavior. To measure the acceptance of new technology, several key components should be identified and explained in the literature on e-banking services, IT and psychology. Lee and Lee (2014) mentioned in their research that understanding human behavior is quite difficult to measure. Empirical research in various contexts suggests that the distal nature of decision-making in adopting or rejecting a product depends on the intention-behavior relationship of humans.

Initially, Fishbein and Ajzen (1975) introduced an additional construct, "Perceived Behavioral control," to the TRA model to cover the intention-behavior gap, but the theory was limited to the field of psychology. In the literature on technological adoption theories,

the TAM and unified theory of acceptance and use of technology (UTAUT) models are the most parsimonious models to explain factors influencing the adoption of new technologies; however, they are unable to address the influence behind changes in human intentions and behavior. [Moghavvemi et al. \(2018\)](#) and [Bhatterjee and Lin \(2015\)](#) defined the intention-behavior gap as the discrepancy between a user's intention for a specific behavior and their actual behavior. The causes and effects of this discrepancy have been described and highlighted in various fields accordingly. Social, personal and environmental aspects are the main reasons behind the discrepancy. [Krosnick et al. \(1993\)](#) and [Holland et al. \(2002\)](#) highlighted attitude as the causal factor in the intention-behavior gap. [Moghavvemi et al. \(2018\)](#) lodged any event which may divert the intention of a person to perform any act, such as migration. The literature reveals that the field of health sciences has the highest number of empirical studies on the intention-behavior gap.

Over time, various studies have been presented, incorporating the intervention of moderators and mediators, to expand the understanding of the intention-behavior gap. [\(Hassan et al., 2016; Grimmer and Miles, 2017\)](#) presented planning as a mediator having a positive and significant impact on the intention-behavior relationship. In addition, [Gollwitzer et al. \(2009\)](#) and [Sheeran \(2005\)](#) divided planning into action and coping planning. [Scholz et al. \(2008\)](#) found planning as the mediator in the health science context. [Sniehotta et al. \(2005\)](#) defined action planning as the process of linking forward intended behavior, such as when, where and how to act, while coping planning is a kind of supportive plan to cope with any barriers toward completion of intended behavior. Broadly, contextual and cultural factors significantly influence the transition from intention to actual usage of online services, such as the low financial literacy rate of the population, which may lead to a lack of consideration of the economic benefits [\(Khan and Arif, 2022\)](#). In Pakistan, a collectivist pattern of decision-making exists, which determines whether or not to adopt a certain service. Therefore, another aim of the current study is to explore planning as a mediator in the uncertain context of Pakistan, where infrastructure issues and demographic factors, along with cultural factors, make it a unique online banking landscape.

Linking above, [Sheeran \(2002\)](#) and [Kuhl's \(1985\)](#) determined the action-control scale of personality by following the studies of and [Schifter and Ajzen \(1985\)](#). They both highlighted personality as a significant moderator in Intention-Behavior relationships. [Maier et al. \(2012\)](#) highlighted in the study that individuals alter intentions into behavior according to their personality and users' personality acts as a moderator in intention-behavior relationships in technology adoption. [Devaraj et al. \(2008\)](#), [McElroy et al. \(2007\)](#) and [Benbasat and Barki \(2003\)](#) illuminated in their studies that there is a need for further studies to solve the riddles behind the intention-behavior gap in the field of technological adoption. Cultural and contextual differences influence an individual's personality and behavior [\(Kuhl, 1985\)](#). Particularly, where customers are early adopters and action-oriented in these circumstances, the probability of actual usage is high; likewise, in developing countries, such as Pakistan, the situation is the opposite, where most of the customers are late adopters [\(State Bank of Pakistan, 2023\)](#), may have a state-oriented personality. Hence, personality affects the degree of acceptance of technology and is therefore considered an important factor to be highlighted through the current study.

2.1.2.2 Integration of expectation continuance theory (ECT) with technology acceptance model (TAM): service quality, satisfaction and continuous usage. The expectation-confirmation theory originated from the research domain of customer research and their psychological characteristics. The theory was first proposed by [Oliver \(1980\)](#), outlining the theoretical aspects of customer satisfaction within the decision-making process related to any product or service. As per ECT, "Consumers' satisfaction is geared up by belief/disbelief

programmed as a result of the determination of performance against the expectation". Thoroughly, there have been several updates and extensions introduced in this theory and the latest version of the theory is proposed by Bhattacharjee (2001), which is known as the ECM. In the context of information systems (IS), the expectation-confirmation model is the most widely adopted and extensively used model extended from ECT. The model predominantly focused on the post-usage or post-adoptive variables to describe usage continuance. In this theory, the post-usage expectation is replaced by the post-PU as a post-acceptance expectation, as recommended by Bhattacharjee *et al.* (2007). Timely, additive external variables were introduced, such as user IT self-efficacy and facilitating conditions, which are related to the skills and abilities of users or employees in IT-based organizations that facilitate end-users through high-quality services. Zhang *et al.* (2015) conducted a study in a similar field on the perception or perceived value of online services by applying ECM. Ghorbanzadeh *et al.* (2020), Kim (2023), Kumari and Biswas (2023), Sreelakshmi and Prathap (2023) and Uzir *et al.* (2021) highlighted the significance of perceived value in driving customer satisfaction. All the mentioned studies demonstrated that the perceived value perceived by customers has a significant contribution in shaping the level of satisfaction for any product or service. In addition, the above studies also emphasized that customer satisfaction depends on what they receive in value for their payment, and service quality is one of the main factors that influence the reputation and value of any product or service. Supporting this, Dangaiso *et al.* (2024) quoted internet banking service quality as a focal construct of perceived value or perceived performance. Furthermore, Zeithaml, *et al.* (2002) operationalized e-service quality as a proxy for ease of use. Herington and Weaven (2009) developed an online banking service quality measurement scale built on four dimensions, and self-efficacy is one of them, even though (Amin, 2016; Ui Haq and Awan, 2020) supported the same argument.

ECT and its modifications are widely used in multidisciplinary research contexts. Still, the most widespread application of its extensions has been observed in the field of marketing and e-services, where they are used to explore behavioral factors such as purchase intentions, satisfaction and loyalty. Lin and Sun (2009) adapted ECT to explore how satisfaction with services may drive customer loyalty. Fu *et al.* (2018) conducted a multi-dimensional study on the usage of public transport from the passenger's perspective and concluded that improved service quality will increase satisfaction and sustain passenger loyalty.

2.2 Empirical studies and hypotheses development

This section is aimed to discuss the relationship between variables in terms of direct and indirect relationship as per highlighted in the previous studies.

2.2.1 Perceived usefulness and intention to use. According to Davis (1989), PU is defined as the "belief of a person that use of technology will augment their job performance to perform a task". Chen *et al.* (2019), Kapoor *et al.* (2014), Tan and Teo (2000) marked PU as a key driver and determinant of intention toward adoption of technological innovations. Furthermore, Al-Somali *et al.* (2009), Alalwan *et al.* (2016), Alsajjan and Dennis (2010) proposed "PU" as a positive stimulus of intention to use m-banking services. Theoretical and empirical studies promote "PU" as one of the most influential drivers of intention to use e-banking services (Kapoor *et al.*, 2014; Chen *et al.*, 2014). Afshan *et al.* (2018) hypothesized and confirmed the strong impact of PU in predicting the intention to use online banking services. Rimenschneider *et al.* (2002), Suh and Han (2002) and Sohail and Shanmughan (2003) found the same results on the significance of PU on internet banking adoption. As Venkatesh and Davis (1996) labeled PU as the strongest determinant of intention to use, and Febriandika *et al.* (2023) and Kaulu *et al.* (2024) supported the same argument. So, this study

hypothesized the relation between PU and the Intention to use online banking services as follows:

H1. PU is the strongest stimulus of Intention to use online banking services.

2.2.2 Perceived ease of use and intention to use. PEOU is defined as a “degree of ease to use or operate any prescribed technology to perform some task” (Venkatesh and Davis, 1996), marking PEOU as the second most significant factor to predict intention to use technological advancements. Mazhar *et al.* (2014) presented it as a vital characteristic of e-business apps. Furthermore, Dai and Palvia (2008) found PEOU as a strong determinant of intention to use m-banking services. Akturan and Tezcan (2012) Found the same results between PEOU and Intention to use m-banking services. Wang *et al.* (2003) commented in their study that PEOU is the direct or indirect influential factor of intention to behave. Chau and Lai (2003) recognized PEOU as an effective contributor to the adoption of internet banking services. Mao *et al.* (2005), Davis *et al.* (1989), Wang *et al.* (2003) found the same contribution of PEOU in the intention to use e-services for different purposes in different fields. Novianto *et al.* (2024) emphasized that PEOU is a positive influencer of intention to adopt e-banking services, as found by Kaulu *et al.* (2024). Therefore, we hypothesized that:

H2. PEOU is a stronger stimulus of Intention to use online banking services.

2.2.3 Perceived risks and intention to use. The possibility of uncertainty associated with using any product or service, which can lead to negative outcomes, is known as perceived risk. (Bauer, 1967) defined perceived risk as “a combination of uncertainty plus seriousness of outcome involved”. Typically, perceived risk is measured using Likert scaling questions to assess the degree of perceived risk associated with a particular product.

Fernandes and Awamleh (2006) idealized financial risk and security risk are associated with online banking services due to the possibility of unauthorized access to customer accounts and the compromise of bank operating systems. By following this, Hanafizadeh and Khedmatgozar (2012) described the role of performance, time, security, financial and privacy risk in using online banking services. In addition, Raj *et al.* (2024) found the negative correlation between perceived risks and purchase intention for online shopping.

2.2.4 Time risk and intention to use. One of the primary benefits of online banking services is the time-saving aspect. Some researchers count it as a time risk (Lee, 2009). Two main drawbacks of online banking services, which can sometimes be time-consuming for customers, include slow website page loading and inappropriate websites, leading to dissatisfied customers with online banking services. Based on the literature review, the following hypothesis is drawn:

H3.1. Time risk is negatively affected by the intention to use online banking services.

2.2.5 Privacy risk and intention to use. Akturan and Tezcan (2012) defined privacy risk as “Risk associated with control over personal information by another person or entity”. Casess (2002) and Pikkarainen *et al.* (2004) proclaimed that the emergence of technological developments on the internet, leading to hyperspace e-services, is directed to evolve new forms of risk, such as privacy risk. Indeed, Finn *et al.* (2013) discussed privacy concerns related to data storage, location and association in the operating systems of service providers, which may be accessed and hacked by a third party. Hence, privacy concerns may deter customers from adopting online banking services. Empirical studies in various contexts have

provided evidence of the significant impact of privacy risk on the purpose or continuity of using online banking services, as noted in studies by AlSheikh and Bojei (2014) and Laukkanen (2007). Last but not least, Kaulu *et al.* (2024) conducted a study on e-banking adoption from a cybercrime perspective and found that privacy risk has no positive influence on e-banking adoption. Hence, theory and empirical studies suggest the following hypothesis:

H3.2. Privacy risk has a negative impact on the intention to use online banking services.

2.2.6 Opportunity cost risk and intention to use. Lu *et al.* (2005) referred to opportunity cost risk as the degree of uncertainty in choosing the best alternative and losing the other options. Kassim and Ramayah (2015) highlighted opportunity cost risk as a facet of the risk dimension related to online banking services. Literature streams on opportunity cost risk have shown that opportunity cost risk is emerging as a second-order dimensional risk that has not been considered in the measurement of risk perceptions, especially in studies from Pakistan. Therefore, we hypothesize opportunity cost risk from the perspective of Pakistan to explore its significance as follows:

H3.3. Opportunity cost risk negatively influences intention to adopt online banking services.

2.2.7 Financial risk and intention to use. Financial risk is considered the “most crucial facet of perceived risk associated with online banking services” (Kassim and Ramayah, 2015). Featherman and Pavlou (2003) and Akturan and Tezcan (2012) re-narrated financial risk as “the potential monetary outlay associated with the initial purchase prices as well as the subsequent maintenance cost of the product”. Kuisma *et al.* (2007) pointed out in a study that most customers are afraid of incurring financial loss while paying any charges or funds transfers via the internet. Entering incorrect information may result in transferring funds to undesired recipients, for which service providers will not be compensated. This outcome acts as a barrier for customers to perform online banking transactions or has low intentions to use online banking services. Hence, we hypothesize as follows:

H3.4. Financial risk negatively influences the intention to adopt online banking services.

2.2.8 Cyber security risk and intention to use. In the era of technological revolution, when innovators are at the forefront of modernizing business inventions, hackers are also active and involved in performing fraudulent acts. Phishing is a familiar tactic used by hackers to acquire sensitive information, such as passwords and debit-credit card codes, from trustworthy accounts via electronic means (Reavley, 2005). It highlights spoof emails as a source of gaining sensitive information from trustworthy phishers. Casalo *et al.* (2007) affirmed security, including the protection of personal data and interrelated issues in the case of dealing with phishers or hackers. On the other hand, the degree of assurance provided by financial institutions is related to the operating systems or technology they use. Distinctive studies have confirmed the role of cyber-security risk in the intention to adopt online banking services, such as Gu *et al.* (2009), Laukkanen (2007); Mashhour and Saleh (2015) and Al-Husein and Sadi, (2015). Linking above, Akinbowale *et al.* (2023) claimed that cybersecurity risk can impact e-banking adoption if the customer feels fearful and insecure in conducting online banking transactions. Therefore, we hypothesize that:

H3.5. Cyber-security risk negatively influences intention to adopt online banking services.

2.2.9 *Subjective norms and intention to use.* Ajzen (1991) defined subjective norms as “the social pressures on the individual to perform or not to perform a particular behavior”. Kalafatis *et al.* (1999) highlighted family, friends and colleagues are the sources of social pressure to act in a specific way. At the same time, Picoto and Pinto (2021) and Szopiński (2016) added cultural norms as an influencing factor to adopt e-banking services. In Islamic countries, people are increasingly turning to Islamic banking services. A study by Ab. Rahim and Amin (2011) explored the subjective norm as a significant driver of customers adopting Islamic insurance. In addition, Abduh *et al.* (2011) found the same factors associated with drawl and depositing activities in Islamic banks. Ghaazi *et al.* (2024) Probed the same from the perspectives of Pakistan and found a positive impact of subjective norms on purchase intention. Hence, it is hypothesized that:

H4. Subjective norms positively influence intention to adopt online banking services.

2.2.10 *Perceived benefits and intention to use.* In a decision-making process, people usually analyze the “cost-benefit” approach to purchasing any product or service (Wright, 1975). In addition, Kim *et al.* (2007) conducted a study on mobile banking services adoption in terms of the hedonic and utilitarian benefits of services behind their adoption. Lee (2009) highlighted low transaction charges, speedy transactions and transparency in displaying information, 24/7 services are the primary benefits provided by banking services for customers (Oh *et al.*, 2008). Moreover, Lee (2009) added high deposit rates, exemption of handling transaction fees and bonus points promotions are benefits for customers that can be enjoyed using online banking services more than traditional banking services. In addition, Kaur and Arora (2023) advocate that perceived benefit has found a significant influence on to adoption of online services. Supporting this, Wang *et al.* (2024) proposed that perceived benefits are one of the crucial motivations behind espousing any technology. Therefore, we hypothesize that:

H5. Perceived benefits positively influence intention to adopt online banking services.

2.2.11 *Intention to use and actual usage.* Intention to use is the basic factor determining an individual’s actual usage of information technology (Osei *et al.*, 2022). Supporting the above, Ain *et al.* (2016) and Mohammadi (2015) found a positive and significant relationship between intention to use and actual usage. Furthermore, Ajzen (1991) proposed that intention is an essential indicator of actual behavior. Pavlou and Fygenson (2006) found the same intervention between Intention to use and actual usage. Therefore, we hypothesized that:

H6. Intention to use has a direct and significant relation with actual usage.

2.2.12 *Planning and actual usage.* According to Pfeffer and Strobach (2019), individuals who usually make plans more often intend to act in an actual way. Gollwitzer (1993) commented that by planning, an individual can target their future intentions; in other words, planned responses can be performed easily. A planned behavior is composed of two types: planning action and coping planning. Scolze *et al.* (2008) found planning to be the most critical factor for behavioral maintenance, a finding also noted by Sniehotta *et al.* (2005). Literature suggests that:

H7. Planning has a direct and significant relation with actual usage.

2.2.13 Personality and actual usage. The adoption of innovative technological advancements hinges on the characteristics of customers, as not all customers behave in the same manner at the same time (Rogers, 1962). Several researchers have contributed to the research stream on online banking adoption and customer perception, especially in developing countries. Supporting that, Pikkarainen *et al.* (2004), Akinci *et al.* (2004) and Poon (2008) stated that a customer's psychological traits such as personality have a strong influence on the usage of any technology. Suppose that the customer's alacrity to bear risk in online banking has been connected to the customer's characteristics Aldás-Manzano *et al.* (2009) and Lassar *et al.* (2005). In addition, Aaker (1997) backed that customers' personality impacts their appraisal of brands, and the selection of products or services depends on the congruency customers identify with the product or services and their personality traits (Khare *et al.*, 2010). In addition, a study was conducted and found that personality traits can be considered a significant predictor of actual usage. In the field of information technology, Malik and Singh (2022) concluded that an individual's personality directly affects the adoption and use of green banking channels in India as well.

MacCann *et al.* (2015) counted the impact of personality traits on health behavior through the Big Five personality model. In the field of information technology, Malik and Singh (2022) measured the indirect effect of personality on Actual usage, but this study will assume the direct impact of personality on actual usage. Several researchers argued for the addition of personality traits in TAM. Urueña *et al.* (2018), highlighting that personality traits give direct or indirect effects in adopting new technologies, such as flexibility and a tolerant nature of the individual. Furthermore, disciplined people usually adopt technology for productive purposes rather than leisure (Xu *et al.*, 2016). In some cases, people who are self-conscious about their personality do not believe they can become innovators or be the first to experience something (Devaraj *et al.*, 2008). In contrast, people who are confident and friendly are more likely to adopt changes and new technology (Agyei *et al.*, 2020). Therefore, we hypothesized that:

H8. Personality has a direct influence on actual usage.

2.2.14 Personality as a moderator between intention to use and actual usage. Hayes (2018) defined moderation as a phenomenal interaction that ensues when a third variable influences the strength of the causal association between the dependent and independent variables. For this tenacity, Personality traits are proposed to moderate the intention-behavior relation and collaborate to describe a user's act on his intention sometimes, whereas sometimes diverge to act as described in the TAM supported (Robey, 1979; Naeem *et al.*, 2023). Supporting this, Ebrahimi *et al.* (2019) quoted that human reactions and decision-making processes are highly affected by the personality traits of users. Unifying the above, literature suggests that personality serves as a moderator between intention to use and actual usage. For instance, Sheeran (2002) conducted a study based on scale and found state-action personality to be a significant moderator. Supporting this, Bagozzi and Warshaw (1990), Gollwitzer (1993) and Jones *et al.* (2011) demonstrated that a person's behavioral characteristic in terms of personality is a powerful tool in determining and implementing an intention into action. Furthermore, Maier *et al.* (2012) conducted a study by taking dispositional resistance as a proxy variable of personality and found a moderation effect, while MacCann *et al.* (2015) found no moderation effect between thrice variables. Shortly, Svendsen *et al.* (2011) and Rebar *et al.* (2019), profound moderation impact of personality between intention and actual behavior in technology acceptance. In addition, Bogg and Roberts (2013) expounded on personality traits that have impulse control over the person's actual intention. Following the above, Rhodes *et al.* (2022) explored in 80% of cases, the

personality factors have acted as a moderator between intention and action in physical activities. [Conner and Norman \(2022\)](#) enumerated personality is one of the moderators of the intention-behavior gap, and recommended further studies to explore more complex results on the moderators to bridge the intention-behavior gap. [Liao et al. \(2023\)](#) tested the hedonic and cognitive factors of behavioral intention and the relationship between intention to behave and attitude, and found customer personal traits to be a significant moderator between behavioral intention factors and attitude toward using e-banking in Cambodia. So, the proposed study assumes that:

H9. Personality acts as a moderator between intention to use and actual usage.

2.2.15 Planning as a mediator between intention to use and actual usage. [Pfeffer and Strobach \(2017\)](#) found intervention of planning between Intention and Behavior. Supporting this, [Scholz et al. \(2008\)](#) and [Norman and Aron \(2003\)](#) identified the mediating role of action-coping planning between intention and actual behavior. Furthermore, [Pfeffer and Strobach \(2021\)](#) and [Reuter et al. \(2008\)](#) claimed a significant mediated effect of planning on intention-behavior. In health-action process approach (HAPA), planning is counted as a substantial mediator [Scholz et al. \(2008\)](#), while [Judd and Kenny \(1981\)](#) claimed the mediation effect is nothing but a correlational phenomenon. [Lippke et al. \(2009\)](#) found the partial mediating effect of planning between intention-behavior in the field of physical sciences. Supporting this, [Sniehotta et al. \(2005\)](#) and [Wiedmann et al. \(2009\)](#) found the same results in the field of health sciences. Literature directs the hypothesis as:

H10. Planning acts as a mediator between intention to use and actual usage.

2.2.16 Actual usage and satisfaction. Actual usage can be defined as the frequency of using technology or e-services ([Isaac et al., 2017](#); [Kim et al., 2009](#)). [Delone and Mclean \(2003\)](#) concluded that the amount of use can affect the degree of satisfaction negatively or positively. Following this, many studies have been conducted on the impact of actual usage on satisfaction, such as [Makokha and Ochieng \(2014\)](#) found a positive influence of actual usage on user satisfaction. Likewise, [Iyanda \(2016\)](#) found the same results in the context of Nigeria for the usage of smartphones. [Chau and Lai \(2003\)](#) highlighted interrelated characteristics between actual usage and satisfaction. Hence, the literature claims that:

H11. Actual usage leads to satisfaction of customers for online banking services.

2.2.17 Satisfaction and continuous usage. The decision to re-use or re-purchase any service or product lies in the level of satisfaction ([Anderson and Srinivasun, 2003](#)). [Yaslioglu et al. \(2013\)](#) enlightened that higher satisfaction yields loyalty from customers, which results in customers' intention to reuse the services. Furthermore, [Yuan et al. \(2018\)](#) and [Li and Shang \(2020\)](#) found the same results and concluded that satisfaction is not a core variable but a significant variable to affect customers' repurchase decision-making for e-services. Kottler and (Armstrong) labeled customer satisfaction as a key to the intention to repurchase in the future. [Rita et al. \(2019\)](#) claimed a significant contribution to satisfaction on the repurchase intention of customers for online shopping. In addition, [Fu et al. \(2023\)](#) contended that enhancing user satisfaction will yield continuity in using any e-service apps. Confirming this, [Yu et al. \(2024\)](#) concluded from a study on an AI chatbot that the user's level of satisfaction is the primary determinant of their willingness to continue using the service. Therefore, we hypothesized that:

H12. Satisfaction has a positive impact on the continuous usage of online banking services.

2.2.18 Service quality and continuous usage. Service quality determines the deviation between feelings and needs, expectations and services received. In addition, [Zeithaml, et al. \(2002\)](#) define e-services quality as “The extent to which a website facilitates efficient and effective shopping, purchasing and delivery of products and services. The significance and direct effect of service quality on customers’ repurchase decisions have been well described within the literature. [Chomvilailuka and Butcher \(2014\)](#), [Kim et al. \(2004\)](#), [Izogo \(2016\)](#) found a linear positive relationship between service quality and repurchase decision. Moreover, [Zhang et al. \(2011\)](#) marked that service quality has a positive impact on repurchase decisions in B2B e-commerce services. Verifying this, [Chatzigeorgiou and Simeli \(2017\)](#), [Liu and Jang \(2009\)](#) and [Mensah and Mensah \(2018\)](#) conducted studies on the service quality of restaurant services and found that service quality is significantly enough to predict customers’ repurchase decisions. Furthermore, [Olaleye et al. \(2021\)](#) evaluated that high-quality e-services will yield a high customer retention rate. In addition, [Ayinaddis et al. \(2023\)](#) and [Yang et al. \(2023\)](#) advocated the positive correlation between service quality and loyalty or continuous usage of online services. [Ashiq and Hussain \(2023\)](#) conducted a study on the relationship between e-service quality and e-loyalty and found a significant and positive relationship between them. Hence, it is hypothesized that:

H13. Service quality has a direct influence on the continuous usage of online banking services.

2.2.19 Service quality as a moderator between satisfaction and continuous usage. [Eggert and Ulaga \(2002\)](#) and [Whittaker et al. \(2007\)](#) argued that perceived value in terms of service quality plays an important role in increasing the level of satisfaction with products and services. [Parasuraman et al. \(1985\)](#), [O’Neill and Palmer \(2004\)](#) and [Thaichon et al. \(2012\)](#) defined service quality as the difference between expectation and perception of the actual delivered goods. Furthermore, [Thaichon et al. \(2012\)](#) counted the skills and professionalism of the technical team or employees as service quality, which contributes to retaining the customer’s loyalty in the long run. Moreover, satisfaction is firmly related to the service quality of any product or service, and a satisfied customer, who experiences high service quality, will continue to use the service. [Blut et al. \(2015\)](#) and [Kitapci et al. \(2014\)](#) argued a positive relationship between Service quality and Customer satisfaction. In addition, [Rita et al. \(2019\)](#) quoted that e-services quality has a positive influence, directly or indirectly, on satisfaction or repurchase intention. [Cronin and Taylor \(1992\)](#) pointed out that perceived service quality leads to satisfaction and that satisfaction will create a positive impact on repurchase decisions. In short, the level of service quality determines the relationship between satisfaction and continuous usage. [Bitner and Hubbert \(1994\)](#) called service quality as an antecedent of satisfaction, which decides the degree of significance between satisfaction and the repurchase decision. Supporting this, [Nguyen and Dai \(2018\)](#) correlated service quality, satisfaction and continuance intention in the same manner. Supporting this, [Saengchai and Jernsittiparsert \(2020\)](#) confirmed in the study of Thailand that service quality moderates the relationship between satisfaction and continuous usage. Hence, literature suggests hypothesizing that:

H14. Service quality moderates the relationship between satisfaction and continuous usage of online banking services.

2.3 Conceptual model

Based on the above-drawn hypotheses, the following model has been proposed to evaluate the Intention-Retention journey of customers for online banking services in Pakistan, as presented in Figure 1.

3. Methodology

The current study used a quantitative research technique to measure the elements that affect the journey of customers from intention to retention of online banking services in the context of Pakistan.

3.1 Measurements

Luarn and Lin (2004) articulated that items used for constructs should be adopted from previous research to ensure the content validity, readability and formatting of items to minimize the chances of inappropriate responses. Hence, the following Table 1 demonstrates individual constructs along with sources:

The present study relies on a five-point Likert scale, ranging from 1. Strongly disagree with 5. Strongly agree. Before data collection, the designed questionnaire was analyzed by experts and underwent pilot testing to ensure that the reliability requirements were met. Following this, the targeted sample size for data collection was determined. By analyzing the results of a pilot study, it was found that the questionnaire was reliable and internally consistent, making it suitable for collecting the required data from prospective respondents.

3.2 Sample profile

The progressive tendencies toward digital literacy and the accessibility of technology are generating sustainable, positive outcomes for emerging economies to adopt e-service facilities in every field of life. Following the 2019 pandemic, most customers have shifted

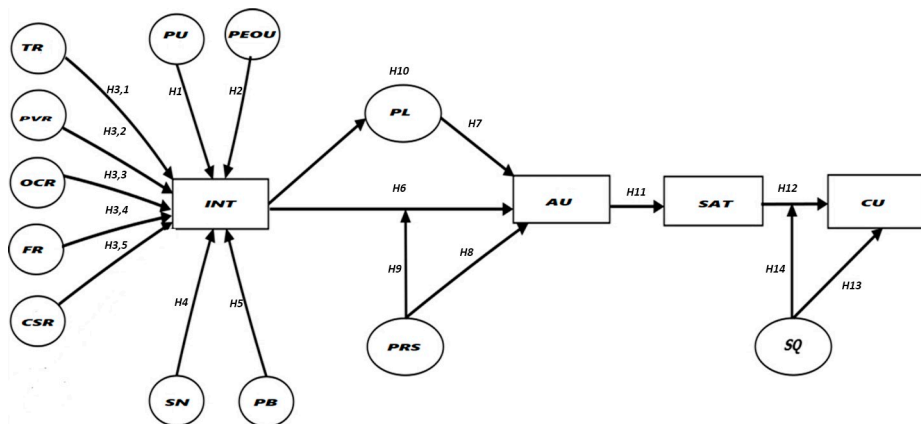


Figure 1. Hypothesized model

Note(s): Where: SN: subjective norms, PU: perceived usefulness, PEOU: perceived ease of use, PB: perceived benefits, TR: time risk= TR, PV: privacy risk, OCR: opportunity cost risk, FR: financial risk, CR: cyber risk, PRS: personality, PL: planning, AU: actual usage, SAT: satisfaction, SQ: service quality, CU: continuous usage

Source: Authors' proposed model

Table 1. Questionnaire constructs and items

Variables	Items	Sources
Intention to use	2	Fathema <i>et al.</i> (2015), Kassim and Ramayah (2015)
Perceived benefits	4	Akturan and Tezcan (2012), Chavali and Kumar (2018)
Perceived ease of use	4	Davis <i>et al.</i> (1989)
Perceived usefulness	3	Kassim and Ramayah (2015)
Subjective norms	2	Bananuka <i>et al.</i> (2019)
Cyber-security risk	2	Baabdullah <i>et al.</i> (2019), Kassim and Ramayah. (2015)
Financial risk	2	Chavali and Kumar (2018), Kassim and Ramayah (2015)
Opportunity-cost risk	2	Kassim and Ramayah (2015)
Privacy risk	2	Chavali and Kumar (2018), Baabdullah <i>et al.</i> (2019)
Planning	3	Sniehotta <i>et al.</i> (2005)
Personality	3	Kuhl (1982)
Actual usage	3	Dumpit and Fernandez (2017), Oertzen and Schröder (2019)
Continuous usage	5	Kang <i>et al.</i> , (2012), Khalifa and Liu (2007), Oertzen and Schröder (2019)
Satisfaction	3	Assegaff (2016, Taylor and Baker (1994)
Service quality	5	Ahangar (2011), Taylor and Baker (1994)
Time risk	1	Akturan and Tezcan (2012)

Source(s): Authors' own creation

toward online banking services. According to the bulletin of the State Bank of Pakistan, the growth rate of online banking transactions in Pakistan recorded a 55% increase. The total population of Internet banking users has reached 4.97 million, as reported in the SBP Payment Systems Review (Q3, 2021). As (Comrey and Lee, 1992) considered sample sizes of 1,000 or more to be excellent for result accuracy. While statistically recommended by Hair *et al.* (2019) to be 10 times the number of items for all constructs in the multivariate technique.

Therefore, the sample size has been selected as per the criteria of Hair *et al.* (2019) and Comrey and Lee (1992). The study comprises a sample size of 2255 from all over Pakistan, spanning various cities. Furthermore, the study approached online banking users from Islamic and Conventional banking customers based on characteristics of subjective norms. The composition of the sample profile has been illustrated in Table 2. The structural model will be analyzed using Partial Least Squares Structural Equation Modeling, as recommended by Hair (2011).

3.2.1 Insights from Islamic versus conventional banking users. In this research, the authors meticulously scrutinized the disparities between customers of Islamic and conventional banks, illuminating their adoption of online banking services from the context of Pakistan along with socio-economic factors. Considering that, most of the sample of this research includes Islamic banking users. The construct of subjective norms represents an in-depth analysis to describe the reinforcement of friends and family on customers' decision-making in the adoption of Islamic or conventional online banking services. The proposed findings of this study reveal an intriguing trend in Pakistan, showing that Islamic banking customers exhibit a higher propensity to use online banking, which suggests that social networks, diversifying demographic factors and family influences are the major causes behind this trend. In addition, including relevant items in the construct of subjective norms can provide a more comprehensive understanding of subjective norms in the Pakistani context. The tactic is predominantly appropriate and verified by Raza *et al.* (2020) in the context of Pakistan, where both Islamic and Conventional banking systems exist.

Table 2, where various demographic factors indirectly enlighten the disparity between modes of conducting online banking services. For instance, the gender disparity in Table 2, with 75.2% of male users, suggests that cultural norms may be influencing the prioritization

Table 2. Descriptive statistics of sample profile

Characteristics	Demographics	Frequency	In %
Age	18–30	720	31.92
	31–42	747	33.12
	43–54	519	23.0
	55 or above	269	11.9
Gender	Male	1696	75.2
	Female	559	24.7
City	Karachi	783	34.7
	Lahore	474	21.01
	Islamabad	569	25.2
	Other	429	19.2
Status	Employed	2099	93.08
	Unemployed	156	6.9
Online banking users	Yes	2255	100
Mode of banking	Islamic	1067	47.31
	Conventional	860	38.1
	Both	328	14.5

Source(s): Authors' own creation

of Islamic banking services over conventional banking services, as men are often the primary decision-makers in financial matters in Pakistan, according to cultural norms. The composition of users includes 47.3% Islamic banking users, as shown in Table 2, which indicates a significant inclination toward Shariah-compliant services driven by religious beliefs and security concerns. Meanwhile, 38.1% are from Conventional banking, driven by familiarity and a broader range of financial services. The smaller group, comprising 14.5% of users, is of both types, advocating a pragmatic approach tailored to their specific needs.

Therefore, this research acknowledges the need for further study directions, specifically comparative studies between the behavioral and psychological traits of customers of Islamic and Conventional banking, as it requires a detailed analysis of the disparities between the two cohorts.

4. Data analysis

The hypothesized model has been investigated by a dual-dimensional methodology standardized by Hair *et al.* (2017, 2021) including outer model assessment and Inner model assessment. Smart PLS-3 software has been used to process and mine the outcomes from data by using statistical scales under the gauge determined by Hair *et al.*'s (2011, 2021) model.

4.1 Outer model measurements

To explore the relationship between a dependent variable and independent variables, a measurement model has been constructed in SmartPLS-3, assessing inner model measurements as directed in previous models. Reliability, consistency and validity are the primary measures used to evaluate outer model measurements.

4.1.1 Reliability testing. CR values between 0.7 and 0.9 ranges are considered satisfactory to good following the recommendations of Hair *et al.* (2018) and Urbach and Ahlemann (2010) CR has been computed and measured. Therefore, CR has been computed and measured in this research to verify the consistency of the variables. Table 3 represents the value of the CR of all variables and the CR value of all variables lies on the acceptance region.

4.1.2 *Convergent validity.* [Hair et al. \(2011, 2013\)](#) emphasized convergent validity to measure intercorrelation between variables. For this purpose, a threshold value of 0.5 or above is restricted ([Henseler et al., 2015](#)). In [Table 3](#), all variables have values greater than 0.5, indicating no intercorrelation among the variables.

After AVE, factor loadings are considered another technique to measure the validity of constructs. Following steps from previous studies, 0.70 or more than the value of items is considered acceptable as per the directions of [Henseler et al. \(2015\)](#) and [Hair et al. \(2017\)](#). Results, as shown in [Table 3](#), indicate that all values are higher than 0.70. All of them are above 0.7, except a few of them, but AVE values are good, so no need to remove those items from assurance and reliability from the model. [Khan et al. \(2019\)](#) and [Jawaid et al. \(2021\)](#) studies also faced this issue, in which a few values are below 0.7, but AVE values are greater than 0.5

4.1.3 *Discriminant validity.* [Table 4](#) presents a correlation matrix illustrating the inter-construct correlations determined by [Hair et al. \(2016\)](#). Fornell and Larcker, as well as cross-loading factors, are the main determinants of discriminant validity [1]. Fornell and Lacker (1981) are traditional metrics that have been suggested by [Hair et al. \(2018\)](#), in which a particular variable should show more variance with its items rather than with the other variables. For this purpose, [Table 4](#) represents the correlation matrix, most of the variance shows a greater value of the square root of AVE than the respective inter-construct correlation ([Hair et al., 2011](#)). In addition, all of the constructs fulfilled the other requirements of internal consistency and construct validity as well, while some values show a higher value than the diagonal, such as (INT vs PEOU and CU vs PL). The observed higher value of INT vs PEOU, in [Table 4](#), showing discriminant validity aligns with the related theoretical framework on technology and its adoption, such as the TAM, task technology fit (TTF) and social cognitive theory (SCT). Furthermore, the empirical studies in the research streams support the concept that customers' perception of ease of use positively affects the intention to use e-services. Consequently, the correlation may be higher than the threshold value, which is empirically sustained by the studies of [Davis \(1989\)](#), [Gefen and Straub \(2005\)](#), [Lu et al. \(2005\)](#), [Venkatesh and Bala \(2008\)](#), [Akturan and Tezcan \(2012\)](#) and [Mazhar et al. \(2014\)](#). [Table 4](#) indicates that almost all of the variables show a greater value of the square root of AVE than respective inter-construct correlations.

Cross-loadings are another tool of measurement for discriminant validity. [Hair et al. \(2011, 2014, 2018, 2021\)](#) recommended that cross-loadings of individual items should be greater than the cross-loadings on another construct. [Gefen and Straub \(2005\)](#) and [Henseler et al. \(2015\)](#) advised a difference of 0.1 between the cross-loading on their own and another construct to scrutinize discriminant validity. [Table 5](#) indicates values of cross-loadings satisfying conditions.

To measure full multicollinearity, authors performed the Variance Inflation Factor (VIF) among constructs in the inner structural model. [Hair et al. \(2019\)](#), [Pallant \(2020\)](#) and [Sarstedt et al. \(2021\)](#) prescribed threshold value of 5, indicating no multicollinearity issues. In [Table 6](#), all the VIF values are lying between 1 and 3.175, which are below the threshold value. For instance, the construct PEOU indicated the highest value of VIF, which is well within the satisfactory limits, which verifies that there is no underlying predictor variance in the research model. So, the VIF values below 5 indicate no issue of discriminant validity.

However, HTMT inference is the most lenient approach, often maintaining discriminant validity even for highly correlated constructs with correlations close to 1.0 ([Franke and Sarstedt, 2019](#)), especially when examining models with large sample sizes or homogeneous high loadings ([Henseler et al., 2015](#)). Due to this range of specificity, the research context and methodological rigor play a crucial role in carefully selecting the criteria. To avoid overly conservative rejections of validity, for example, HTMT.90 or HTMT inference may

Table 3. Reliability testing and convergent validity

Construct	Items	Loadings	<i>p</i> -values	CR	AVE
AU	AU1	0.920	0.000	0.943	0.847
	AU2	0.938	0.000		
	AU3	0.902	0.000		
CSR	CSR1	0.978	0.000	0.978	0.956
	CSR2	0.977	0.000		
CU	CU1	0.926	0.000	0.935	0.745
	CU2	0.724	0.000		
	CU3	0.895	0.000		
	CU4	0.884	0.000		
	CU5	0.871	0.000		
FR	FR1	0.987	0.000	0.889	0.802
	FR2	0.793	0.000		
INT	INT1	0.961	0.000	0.953	0.911
	INT2	0.948	0.000		
OCR	OCR1	0.903	0.000	0.805	0.676
	OCR2	0.732	0.000		
PB	PB1	0.955	0.000	0.976	0.911
	PB2	0.917	0.000		
	PB3	0.984	0.000		
	PB4	0.960	0.000		
PEOU	PEOU1	0.899	0.000	0.866	0.621
	PEOU2	0.779	0.000		
	PEOU3	0.681	0.000		
	PEOU4	0.778	0.000		
PL	PL1	0.932	0.000	0.919	0.793
	PL2	0.921	0.000		
	PL3	0.813	0.000		
PRS	PRS1	0.581	0.000	0.822	0.614
	PRS2	0.899	0.000		
	PRS3	0.834	0.000		
PU	PU1	0.768	0.000	0.920	0.794
	PU2	0.953	0.000		
	PU3	0.940	0.000		
PVR	PVR1	0.960	0.000	0.768	0.636
	PVR2	0.592	0.000		
SAT	SAT1	0.944	0.000	0.912	0.777
	SAT2	0.899	0.000		
	SAT3	0.795	0.000		
SN	SN1	0.918	0.000	0.818	0.694
	SN2	0.738	0.000		
SQ	SQ1	0.790	0.000	0.922	0.705
	SQ2	0.842	0.000		
	SQ3	0.708	0.000		
	SQ4	0.937	0.000		
	SQ5	0.902	0.000		
TR	TR1	1.000	0.000	1.000	1.000

Source(s): Authors' estimations

Table 4. Fornell–Larcker criterion

	AU	CU	CSR	FR	ITU	OCR	PB	PRS	PL	PVR	SAT	SQ	SN	TR	PEOU	PU
AU	0.92															
CU	0.682	0.863														
CSR	-0.6	-0.335	0.978													
FR	-0.597	-0.776	0.751	0.895												
ITU	0.949	0.57	-0.612	-0.515	0.954											
OCR	-0.173	-0.164	0.281	0.265	-0.216	0.822										
PB	0.18	0.137	-0.178	-0.16	0.159	-0.371	0.954									
PRS	-0.191	-0.045	0.191	0.028	-0.259	0.298	-0.381	0.783								
PL	0.822	0.873	-0.356	-0.613	0.703	-0.097	0.156	-0.046	0.89							
PVR	-0.009	-0.004	0.0213	0.013	-0.007	-0.001	0.003	-0.001	-0.005	0.797						
SAT	0.901	0.773	-0.551	-0.664	0.827	-0.199	0.183	-0.102	0.812	-0.001	0.881					
SQ	0.827	0.793	-0.629	-0.791	0.839	-0.219	0.135	-0.198	0.767	-0.007	0.756	0.84				
SN	0.248	0.183	-0.104	-0.112	0.269	-0.382	0.713	-0.508	0.18	0.004	0.233	0.22	0.833			
TR	-0.784	-0.508	0.586	0.419	-0.759	0.241	-0.175	0.245	-0.567	0.011	-0.814	-0.513	-0.237	1.00		
PEOU	0.661	0.585	-0.495	-0.509	0.748	-0.179	0.158	-0.345	0.473	-0.012	0.615	0.703	0.236	-0.623	0.788	
PU	0.724	0.577	-0.401	-0.463	0.778	-0.08	0.145	-0.208	0.598	0.004	0.781	0.733	0.214	-0.515	0.777	0.891

Note(s): The italic values in diagonal represents the square root of Average Variance Extracted of each construct
Source(s): Authors' estimations

Table 5. Cross-loading factors

	AU	CSR	CU	FR	ITU	OCR	PB	PEOU	PL	PRS	PU	PVR	SAT	SN	SQ	TR
AU1	0.921	-0.400	0.555	-0.415	0.825	-0.134	0.150	0.498	0.705	-0.122	0.503	-0.009	0.788	0.231	0.611	-0.739
AU2	0.938	-0.674	0.811	-0.767	0.859	-0.213	0.191	0.705	0.811	-0.189	0.720	-0.010	0.942	0.238	0.857	-0.819
AU3	0.902	-0.570	0.500	-0.445	0.940	-0.127	0.155	0.613	0.749	-0.215	0.769	-0.006	0.749	0.217	0.803	-0.603
CSR1	-0.601	0.979	-0.266	0.708	-0.608	0.273	-0.166	-0.435	-0.288	0.174	-0.412	0.017	-0.569	-0.106	-0.591	0.587
CSR2	-0.575	0.978	-0.392	0.764	-0.589	0.277	-0.184	-0.537	-0.412	0.201	-0.374	0.025	-0.508	-0.098	-0.642	0.560
CU1	0.567	-0.322	0.926	-0.742	0.466	-0.239	0.126	0.422	0.802	0.059	0.432	-0.001	0.748	0.158	0.646	-0.525
CU2	0.416	-0.078	0.725	-0.485	0.241	0.060	0.071	0.235	0.616	-0.007	0.503	0.008	0.594	0.086	0.501	-0.262
CU3	0.643	-0.179	0.896	-0.615	0.513	-0.170	0.104	0.471	0.729	-0.011	0.500	0.003	0.800	0.206	0.624	-0.597
CU4	0.617	-0.300	0.884	-0.661	0.602	-0.194	0.126	0.716	0.776	-0.113	0.516	-0.011	0.588	0.194	0.765	-0.426
CU5	0.669	-0.517	0.871	-0.814	0.592	-0.127	0.156	0.630	0.829	-0.115	0.549	-0.015	0.607	0.137	0.851	-0.358
FR1	-0.660	0.726	-0.850	0.987	-0.577	0.269	-0.163	-0.563	-0.698	0.065	-0.496	0.013	-0.719	-0.142	-0.839	0.502
FR2	-0.206	0.658	-0.303	0.794	-0.152	0.187	-0.109	-0.179	-0.139	-0.115	-0.223	0.010	-0.288	0.027	-0.411	0.000
INT1	0.957	-0.554	0.638	-0.514	0.962	-0.158	0.152	0.640	0.818	-0.226	0.776	-0.004	0.851	0.250	0.861	-0.716
INT2	0.851	-0.621	0.437	-0.467	0.948	-0.263	0.153	0.803	0.502	-0.275	0.707	-0.010	0.721	0.266	0.733	-0.737
OCR1	-0.170	0.241	-0.154	0.224	-0.211	0.903	-0.385	-0.178	-0.103	0.301	-0.073	-0.002	-0.176	-0.393	-0.207	0.226
OCR2	-0.106	0.227	-0.111	0.221	-0.133	0.733	-0.194	-0.107	-0.047	0.168	-0.058	0.000	-0.152	-0.205	-0.148	0.164
PB1	0.153	-0.138	0.106	-0.117	0.140	-0.371	0.956	0.129	0.121	-0.388	0.107	-0.003	0.154	0.763	0.103	-0.164
PB2	0.014	-0.097	-0.035	-0.025	0.021	-0.367	0.918	0.032	-0.030	-0.406	0.033	-0.001	0.037	0.745	-0.014	-0.070
PB3	0.238	-0.230	0.189	-0.223	0.203	-0.364	0.984	0.174	0.212	-0.357	0.173	0.004	0.235	0.664	0.198	-0.207
PB4	0.115	-0.124	0.095	-0.104	0.108	-0.335	0.960	0.170	0.112	-0.360	0.147	0.009	0.129	0.634	0.070	-0.124
PEOU1	0.466	-0.572	0.435	-0.512	0.544	-0.109	0.143	0.899	0.289	-0.385	0.647	-0.018	0.397	0.153	0.620	-0.433
PEOU2	0.473	-0.185	0.258	-0.127	0.507	-0.057	0.127	0.779	0.215	-0.271	0.420	-0.018	0.324	0.177	0.268	-0.485
PEOU3	0.282	-0.178	0.252	-0.208	0.420	0.008	0.091	0.681	0.275	-0.182	0.633	-0.007	0.174	0.089	0.434	0.002
PEOU4	0.726	-0.516	0.731	-0.612	0.773	-0.301	0.130	0.778	0.591	-0.244	0.704	-0.002	0.825	0.271	0.765	-0.807
PL1	0.842	-0.384	0.865	-0.605	0.712	-0.129	0.167	0.505	0.932	-0.066	0.654	-0.002	0.933	0.208	0.707	-0.746
PL2	0.665	-0.353	0.827	-0.577	0.554	-0.094	0.178	0.506	0.921	-0.043	0.521	-0.010	0.712	0.138	0.608	-0.559
PL3	0.665	-0.203	0.627	-0.447	0.594	-0.027	0.068	0.239	0.814	-0.009	0.396	-0.003	0.478	0.125	0.731	-0.157
PRS1	-0.031	-0.120	-0.059	-0.162	-0.073	0.069	-0.374	-0.260	-0.062	0.592	-0.129	-0.003	0.036	-0.455	-0.049	0.094

(continued)

Table 5. Continued

	AU	CSR	CU	FR	ITU	OCR	PB	PEOU	PL	PRS	PU	PVR	SAT	SN	SQ	TR
PRS2	-0.185	0.174	-0.025	0.004	-0.265	0.284	-0.425	-0.363	-0.053	0.899	-0.197	0.001	-0.046	-0.478	-0.204	0.187
PRS3	-0.160	0.201	-0.051	0.083	-0.195	0.251	-0.195	-0.216	-0.018	0.835	-0.162	-0.003	-0.157	-0.369	-0.150	0.254
PU1	0.434	-0.245	0.609	-0.439	0.485	-0.125	0.102	0.460	0.607	-0.023	0.769	0.013	0.664	0.138	0.564	-0.357
PU2	0.770	-0.509	0.530	-0.471	0.777	-0.049	0.176	0.777	0.581	-0.255	0.953	-0.003	0.795	0.201	0.687	-0.615
PU3	0.679	-0.291	0.461	-0.355	0.770	-0.064	0.107	0.778	0.458	-0.227	0.941	0.005	0.645	0.223	0.702	-0.386
PVR1	-0.009	0.019	-0.004	0.012	-0.008	0.000	0.002	-0.010	-0.005	-0.003	0.003	0.960	-0.002	0.003	-0.007	0.010
PVR2	-0.006	0.017	-0.004	0.011	-0.003	-0.004	0.006	-0.013	-0.003	0.002	0.006	0.592	0.002	0.007	-0.004	0.008
SAT1	0.868	-0.744	0.759	-0.825	0.800	-0.269	0.187	0.550	0.782	-0.078	0.675	-0.006	0.945	0.201	0.817	-0.756
SAT2	0.882	-0.320	0.730	-0.422	0.862	-0.186	0.138	0.611	0.831	-0.147	0.704	0.002	0.899	0.273	0.711	-0.807
SAT3	0.589	-0.367	0.524	-0.496	0.461	-0.030	0.165	0.452	0.478	-0.028	0.711	0.003	0.795	0.126	0.405	-0.559
SN1	0.253	-0.173	0.184	-0.148	0.271	-0.371	0.707	0.222	0.198	-0.321	0.200	0.005	0.256	0.918	0.212	-0.276
SN2	0.142	0.054	0.111	-0.007	0.159	-0.250	0.440	0.167	0.077	-0.624	0.153	0.003	0.104	0.739	0.145	-0.078
SQ1	0.491	-0.622	0.663	-0.828	0.514	-0.259	0.101	0.420	0.534	-0.029	0.615	0.003	0.648	0.122	0.790	-0.317
SQ2	0.680	-0.422	0.647	-0.547	0.771	-0.204	0.107	0.776	0.643	-0.235	0.588	-0.012	0.498	0.217	0.842	-0.387
SQ3	0.635	-0.602	0.177	-0.348	0.716	-0.087	0.072	0.427	0.347	-0.299	0.507	-0.009	0.374	0.144	0.709	-0.309
SQ4	0.803	-0.608	0.607	-0.614	0.855	-0.169	0.121	0.702	0.675	-0.283	0.654	-0.012	0.594	0.211	0.937	-0.458
SQ5	0.852	-0.511	0.876	-0.782	0.751	-0.160	0.140	0.587	0.822	-0.128	0.682	-0.004	0.858	0.214	0.903	-0.592
TR1	-0.785	0.587	-0.509	0.419	-0.760	0.242	-0.175	-0.623	-0.567	0.246	-0.516	0.011	-0.814	-0.238	-0.514	1.000

Source(s): Authors' estimations

Table 6. VIF collinearity

Construct	AU	CU	ITU	PL	SAT
AU					1.000
CU					
CSR			3.010		
FR			2.619		
ITU	2.589			1.000	
OCR			1.320		
PB			2.145		
PRS	1.217				
PL	2.094				
PVR			1.001		
SAT		3.064			
SQ		2.340			
SN			2.276		
TR			2.100		
PEOU			3.175		
PU			2.650		

Source(s): Authors' estimations

be more appropriate in TAMs that examine intention versus actual use constructs, which are conceptually distinct but potentially empirically proximate (Henseler *et al.*, 2015). The current findings support this contingent approach, as nearly all HTMT values in Table 7 fall below their respective thresholds, indicating adequate discriminant validity across all specified criteria.

4.2 Inner model measurement and hypothesis testing

Following the criteria of Henseler (2020) and Hair *et al.* (2018) for outer-model measurements, sub-samples of 5,000 or more have been drawn from the original data through a bootstrapping approach by SmartPLS-3. Values of coefficient, acceptance criteria of hypothesized variables and criterion of R-R-squared and Q-squared are the leading indicators of inter-model measurement as recommended by Haenlein and Kaplan (2004) and Hair *et al.* (2014, 2010).

4.2.1 Predictive relevance of the model. R-squared (coefficient of determination) is one of the basic measurements to find the inner significance of the model. Researchers determined the range of the R-squared value to identify acceptable and unacceptable variance. ($R^2 < 0.19$ = unacceptable) ($0.19 < R^2 < 0.33$ = weak), ($0.33 < R^2 < 0.67$ = moderate) and ($R^2 < 0.67$ = substantial). Table 8 presents the values of R^2 for almost all variables of substantial relevance to the model.

4.2.2 Hypothesis testing. The hypothesized model is designed to analyze the intention-retention cycle of customers for online banking services in Pakistan. For data collection, 2255 samples were drawn from Islamic and Conventional banking users. 19 hypotheses have been drawn to attain the objective of the study. The PLS-SEM technique has been used to examine the significance of the study's hypothesis. For the constructed models, all hypotheses are tested through a three-step moderation analysis, mediation analysis and direct relationship between the dependent and independent variables.

4.2.2.1 Moderation analysis. To find whether personality (PRS) moderates the relationship between intention to use (ITU) and actual usage (AU), this study followed the

Table 7. Heterotrait-Monotrait (HTMT) inferences

Effects	Sample mean (M) [Lower CI; Upper CI]		Sample mean (M) [Lower CI; Upper CI]
CU → AU	0.736 [0.706; 0.766]	SN → CU	0.238 [0.189; 0.282]
CSR → AU	0.638 [0.620; 0.658]	SN → CSR	0.181 [0.147; 0.216]
CSR → CU	0.349 [0.318; 0.379]	SN → FR	0.148 [0.118; 0.182]
FR → AU	0.540 [0.517; 0.565]	SN → ITU	0.352 [0.306; 0.399]
FR → CU	0.721 [0.706; 0.738]	SN → OCR	0.619 [0.562; 0.674]
FR → CSR	0.856 [0.833; 0.878]	SN → PB	0.933 [0.903; 0.964]
ITU → AU	1.043 [1.039; 1.048]	SN → PRS	0.915 [0.874; 0.958]
ITU → CU	0.610 [0.571; 0.655]	SN → PL	0.229 [0.191; 0.273]
ITU → CSR	0.662 [0.643; 0.681]	SN → PVR	0.048 [0.018; 0.092]
ITU → FR	0.464 [0.437; 0.496]	SN → SAT	0.293 [0.243; 0.342]
OCR → AU	0.235 [0.180; 0.290]	SN → SQ	0.288 [0.245; 0.333]
OCR → CU	0.254 [0.214; 0.293]	TR → AU	0.820 [0.804; 0.836]
OCR → CSR	0.394 [0.339; 0.448]	TR → CU	0.528 [0.485; 0.567]
OCR → FR	0.384 [0.322; 0.438]	TR → CSR	0.600 [0.575; 0.623]
OCR → ITU	0.304 [0.249; 0.354]	TR → FR	0.304 [0.281; 0.328]
PB → AU	0.145 [0.114; 0.183]	TR → ITU	0.800 [0.779; 0.822]
PB → CU	0.117 [0.097; 0.142]	TR → OCR	0.321 [0.264; 0.380]
PB → CSR	0.158 [0.117; 0.204]	TR → PB	0.148 [0.109; 0.192]
PB → FR	0.131 [0.090; 0.174]	TR → PRS	0.261 [0.221; 0.306]
PB → ITU	0.132 [0.096; 0.174]	TR → PL	0.588 [0.558; 0.617]
PB → OCR	0.487 [0.439; 0.533]	TR → PVR	0.033 [0.005; 0.076]
PRS → AU	0.198 [0.162; 0.233]	TR → SAT	0.866 [0.853; 0.879]
PRS → CU	0.138 [0.121; 0.158]	TR → SQ	0.515 [0.480; 0.549]
PRS → CSR	0.248 [0.220; 0.277]	TR → SN	0.274 [0.226; 0.329]
PRS → FR	0.220 [0.194; 0.247]	PEOU → AU	0.716 [0.698; 0.732]
PRS → ITU	0.275 [0.239; 0.311]	PEOU → CU	0.636 [0.616; 0.656]
PRS → OCR	0.404 [0.360; 0.442]	PEOU → CSR	0.525 [0.496; 0.556]
PRS → PB	0.513 [0.480; 0.544]	PEOU → FR	0.485 [0.464; 0.505]
PL → AU	0.914 [0.903; 0.924]	PEOU → ITU	0.841 [0.829; 0.854]
PL → CU	0.973 [0.964; 0.983]	PEOU → OCR	0.234 [0.195; 0.278]
PL → CSR	0.389 [0.356; 0.422]	PEOU → PB	0.148 [0.105; 0.194]
PL → FR	0.541 [0.504; 0.584]	PEOU → PRS	0.461 [0.425; 0.505]
PL → ITU	0.773 [0.749; 0.797]	PEOU → PL	0.517 [0.491; 0.544]
PL → OCR	0.139 [0.097; 0.184]	PEOU → PVR	0.048 [0.021; 0.087]
PL → PB	0.139 [0.118; 0.169]	PEOU → SAT	0.651 [0.628; 0.671]
PL → PRS	0.099 [0.076; 0.127]	PEOU → SQ	0.799 [0.786; 0.812]
PVR → AU	0.036 [0.011; 0.077]	PEOU → SN	0.313 [0.256; 0.366]
PVR → CU	0.038 [0.015; 0.074]	PEOU → TR	0.611 [0.593; 0.631]
PVR → CSR	0.040 [0.009; 0.091]	PU → AU	0.787 [0.766; 0.807]
PVR → FR	0.039 [0.012; 0.088]	PU → CU	0.676 [0.648; 0.702]
PVR → ITU	0.034 [0.008; 0.075]	PU → CSR	0.429 [0.401; 0.457]
PVR → OCR	0.048 [0.015; 0.093]	PU → FR	0.474 [0.438; 0.512]
PVR → PB	0.031 [0.010; 0.065]	PU → ITU	0.853 [0.835; 0.871]
PVR → PRS	0.043 [0.018; 0.077]	PU → OCR	0.129 [0.076; 0.189]
PVR → PL	0.037 [0.015; 0.078]	PU → PB	0.130 [0.093; 0.167]
SAT → AU	0.996 [0.990; 1.003]	PU → PRS	0.238 [0.209; 0.267]
SAT → CU	0.864 [0.846; 0.882]	PU → PL	0.699 [0.676; 0.722]
SAT → CSR	0.597 [0.579; 0.614]	PU → PVR	0.037 [0.013; 0.071]
SAT → FR	0.698 [0.688; 0.707]	PU → SAT	0.919 [0.911; 0.928]
SAT → ITU	0.906 [0.885; 0.926]	PU → SQ	0.805 [0.780; 0.827]

(continued)

Table 7. Continued

Effects	Sample mean (M) [Lower CI; Upper CI]	Sample mean (M) [Lower CI; Upper CI]	
SAT → OCR	0.275 [0.228; 0.322]	PU → SN	0.286 [0.239; 0.334]
SAT → PB	0.163 [0.128; 0.202]	PU → TR	0.545 [0.519; 0.570]
SAT → PRS	0.180 [0.158; 0.201]	PU → PEOU	0.889 [0.876; 0.900]
SAT → PL	0.896 [0.881; 0.908]	SQ → ITU	0.942 [0.931; 0.953]
SAT → PVR	0.036 [0.013; 0.075]	SQ → OCR	0.290 [0.237; 0.342]
SQ → AU	0.902 [0.889; 0.913]	SQ → PB	0.106 [0.081; 0.135]
SQ → CU	0.773 [0.750; 0.800]	SQ → PRS	0.283 [0.256; 0.310]
SQ → CSR	0.707 [0.689; 0.724]	SQ → PL	0.810 [0.789; 0.831]
SQ → FR	0.744 [0.723; 0.765]	SQ → PVR	0.039 [0.017; 0.072]
SQ → ITU	0.942 [0.931; 0.953]	SQ → SAT	0.770 [0.752; 0.790]
SQ → OCR	0.290 [0.237; 0.342]	SN → AU	0.321 [0.271; 0.372]

Source(s): Authors’ estimations

Table 8. Coefficient of determination (R^2)

Construct	R square	R square adjusted
Actual usage	0.951	0.951
Continuous usage	0.708	0.708
Intention to use	0.812	0.811
Planning	0.495	0.494
Satisfaction	0.812	0.812

Source(s): Authors' estimations

procedure of (Hair *et al.*, 2017; Baron and Kenny, 1986) to compare the values of R^2 , adjusted R^2 and effect size (f^2). Moreover, service quality has been included as an additional moderator between satisfaction and continuous usage. Using the algorithm, the value of (f^2) has been computed and presented in Table 9. Results indicate that PRS acts as a significant moderator with a small effect size between ITU and AU in the hypothesized model. Secondly, service quality has a small effect according to the criterion and statistically serves as a significant moderator, impacting both satisfaction and continuous usage simultaneously, as presented in Table 9.

4.2.2.2 Mediation analysis. Indirect effects, variance accounted for and type of mediation effects are the determinants of mediation analysis through bootstrapping. The mediation effect between planning, intention to use and actual usage can be seen in Table 10, which shows the indirect effect between planning (PL), ITU and AU. Table 10 shows values of specific indirect effect, such as ($\beta=0.207$, $p=0.000$), which is statistically significant, confirming the mediation effect of planning between intention to use and actual usage. On the other hand, the value of total effects, such as ($\beta=0.971$, $p=0.000$), is statistically significant as well. To determine the effect size of mediation, the present study adopted recommendations by Hair *et al.* (2013, 2014, 2016). In Table 10, the values of specific indirect effects and total effects are presented to determine the type of mediation. According to the recommendations of Hair *et al.* (2021), both statements are statistically significant,

Table 9. Effect size (f^2)

Effects	f^2	Decision
Personality $\times$ intention to use	0.024	Small
Satisfaction $\times$ service quality	0.028	Small
Source(s): Authors' Estimations		

Table 10. Mediation effect: bootstrapped total and indirect effect estimates

Effects	Sample mean (M)	Standard deviation	t-statistics	p-values	Decision	Type
Specific indirect effects ITU → PL → AU	0.207	0.006	36.754	0.000	Significant	Partial
Total effects INT → AU	0.971	0.002	391.362	0.000	Significant	
Note(s): If the <i>p</i> -values of the individual indirect effects and the total effects in the mediation analysis are smaller than the selected significance level (often 0.05), <i>t</i> -statistics ≥1.96, then the effects are deemed significant. The interaction between the independent variable (ITU) and the dependent variable (AU) is only partially accounted for by the proposed mediator (PL), leading to a “Partial” decision in the Type column						
Source(s): Authors’ estimations						

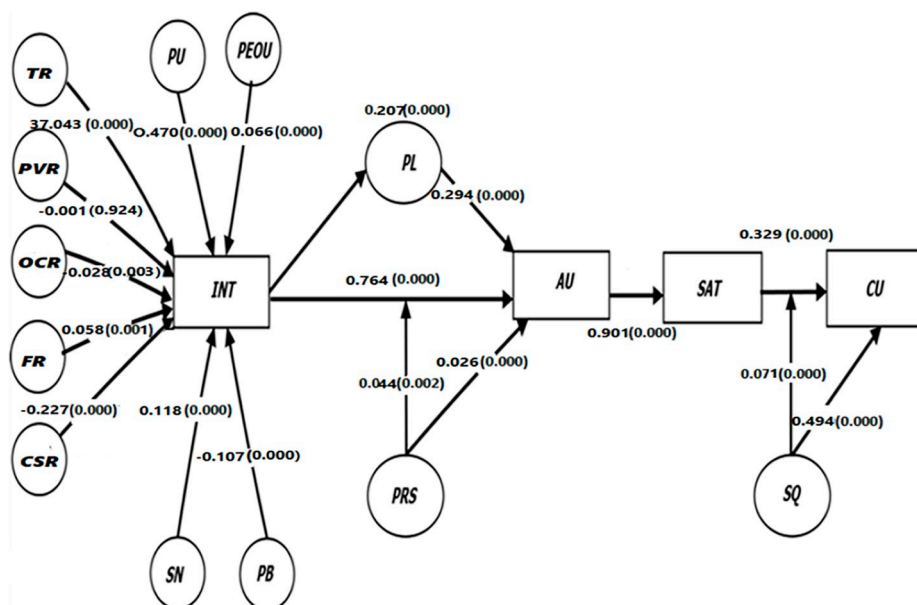
indicating a partial mediation effect of the variable planning on the relationship between intention to use and actual usage.

According to the hypothesized model, this study examined hypotheses in three steps: moderation analysis, Mediation analysis and the direct relationship between dependent and independent variables. Figure 2 has been designed to describe the coefficients and p-values of each variable as follows:

The results are presented in Table 11 to summarize the decision on all drawn hypotheses. Results indicate that PEOU ($\beta = 0.066, p < 0.05$), PU ($\beta = 0.47, p < 0.05$) and SN ($\beta = 0.118, p < 0.05$), PB ($\beta = -0.107, p < 0.05$) are the significant variable to predict ITU online banking services in Pakistan same as the studies of Afshan *et al.* (2018), Ashrafi *et al.* (2014), Siyal *et al.* (2019) and Hussain *et al.* (2020). Therefore, H1, H2, H4, H5 and are supported.

On the other hand, the multi-dimensional risk approach has been tested through the hypothesis drawn. In the penta-dimensional risk approach, time risk (TR) ($\beta = 37.043, p < 0.05$), opportunity cost risk (OCR) ($\beta = -0.028, p < 0.05$), financial risk (FR) ($\beta = 0.058, p < 0.01$), cyber security risk (CSR) ($\beta = -0.227, p < 0.01$) are significant. In contrast, only PVR ($\beta = -0.001, p > 0.05$) is negatively but insignificantly associated with INT, as the follow-up studies of Kassim and Ramayah (2015), Afshan *et al.* (2018) and Akturan and Tezcan (2012). Hence, H3.1, H3.2, H3.4 and H3.5 are supported while H3.2 is unsupported for the hypothesized model of this study.

To bridge the gap between intention-behavior theory, PRS and PL, have been introduced as moderators and mediators, respectively, to predict AU of online banking services in Pakistan. PL ($\beta = 0.294, p < 0.05$) and INT ($\beta = 0.764, p < 0.01$) have a positive and direct influence on AU, which supports H6 and H7, while PRS ($\beta = 0.026, p > 0.05$) has been found as a significant variable as well as directly affecting AU, supporting H8. In indirect effects, PRS ($\beta = 0.044, p < 0.05$) is an essential moderator between INT and AU and directs to

Figure 2. Path coefficient and *p*-values

Source: Authors' own creation

Table 11. Hypothesis testing: partial least squares – structural equation modeling result

No.	Hypothesis	Estimates	Mean	Standard deviation	<i>t</i> -values	<i>p</i> -values	Decision
H1	PU → INT	0.470	0.470	0.013	37.043	0.000	Supported
H2	PEOU → INT	0.066	0.067	0.011	6.044	0.000	Supported
H3.1	TR → INT	37.043	0.000	0.000	0.000	0.000	Supported
H3.2	PVR → INT	-0.001	-0.001	0.009	0.096	0.924	Unsupported
H3.3	OCR → INT	-0.028	-0.029	0.009	3.032	0.003	Supported
H3.4	FR → INT	0.058	0.057	0.017	3.342	0.001	Supported
H3.5	CSR → INT	-0.227	-0.226	0.019	12.009	0.000	Supported
H4	SN → INT	0.118	0.116	0.013	8.789	0.000	Supported
H5	PB → INT	-0.107	-0.105	0.013	7.977	0.000	Supported
H6	INT → AU	0.764	0.764	0.007	114.745	0.000	Supported
H7	PL → AU	0.294	0.294	0.006	45.767	0.000	Supported
H8	PRS → AU	0.026	0.026	0.006	4.456	0.000	Supported
H9	PRS*INT → AU	0.044	0.044	0.005	8.497	0.000	Supported
H10	ITU → PL → AU	0.207	0.207	0.006	36.754	0.000	Supported
H11	AU → SAT	0.901	0.901	0.004	230.420	0.000	Supported
H12	SAT → CU	0.329	0.329	0.016	20.686	0.000	Supported
H13	SQ → CU	0.494	0.494	0.017	29.738	0.000	Supported
H14	SQ*SAT → CU	0.071	0.070	0.006	11.601	0.000	Supported

Note(s): The decision column indicates whether each hypothesis is supported based on the chosen significance level (usually 0.05) for the *p*-value. All hypotheses with *p*-values less than 0.05 are considered supported, while those with *p*-values greater than or equal to 0.05 are considered unsupported

Source(s): Authors' estimations

support *H₉*. Withal, PL ($\beta=0.207, p<0.05$) as a mediator, is significant to mediate the relationship between INT and AU following studies (Scholz *et al.*, 2008; Wiedmann *et al.*, 2009) which support *H₁₀*.

Last but not least, AU ($\beta=0.901, p<0.05$) is found significant to determine level of satisfaction, SAT ($\beta=0.329, p<0.05$) and service quality ($\beta=0.494, p<0.05$) are the significant variables to predict continuous usage of online banking services in Pakistan. Therefore, *H₁₁*, *H₁₂* and *H₁₃* are tremendous variables to predict the dependent variable in this study. Furthermore, service quality as a moderator has statistical significance ($\beta=0.071, p<0.05$), supporting the acceptance of *H₁₄*. Hence, it can be concluded that Service quality moderates the relationship between Satisfaction and Continuous usage.

Multi-group analysis has been recommended to measure significant differences between two groups in a research model. Hence, this study explored the statistical differences for the Intention-retention journey of online banking in Pakistan, likewise the studies of Jawaid *et al.* (2021) and Roberts-Lombard and Petzer (2024). The results are presented in Table 12, which shows that there is a significant variation in financial risk, opportunity cost risk and PU in predicting Intention to use in the group-specific path coefficient between Islamic and conventional banking customers. Proceeding above, satisfaction and service quality also indicate significant variation in predicting continuous usage in the group-specific paths. The reason for the difference may include differences in customers' perceptions of products and services for both groups, such as Islamic banking customers, who are not profit-oriented and therefore are less sensitive to financial risk. Furthermore, religious norms are another main reason why Islamic banking customers do not incur opportunity costs, as their family or colleagues usually influence them to use Islamic banking services. Linking above, service quality is another factor that may matter in front of Islamic banking customers,' Shariah-

Table 12. PLS-multi-group analysis

Hypothesis	Path coefficients-diff	<i>p</i> -value
AU → SAT	-0.014	0.199
CSR → ITU	-0.052	0.195
FR → ITU	0.091	<i>0.012</i>
ITU → AU	-0.004	0.784
ITU → PL	-0.023	0.357
OCR → ITU	-0.054	<i>0.016</i>
PB → ITU	-0.009	0.753
PEOU → ITU	-0.014	0.560
PU → ITU	0.055	<i>0.044</i>
PRS → AU	0.011	0.482
PL → AU	0.011	0.510
PVR → ITU	0.003	0.906
SAT → CU	0.075	<i>0.027</i>
SQ → CU	-0.082	<i>0.020</i>
SN → ITU	0.013	0.658
TR → ITU	0.037	0.306
PRS*INT → AU	0.020	0.153
SQ*SAT → CU	-0.002	0.915

Note(s): The decision column indicates whether each hypothesis is supported based on the chosen significance level (usually 0.05) for the *p*-value. All hypotheses with *p*-values less than 0.05 are considered significant, while those with *p*-values greater than or equal to 0.05 are considered insignificant; the italic values indicate significant values of *p*

Source(s): Authors' estimations

compliant. At the same time, Satisfaction has been considered a secondary priority for Islamic banking users compared to Shariah. The behavioral differences between Islamic and Conventional banking users in adopting digital finance services are witnessed as well in the studies of [Kharrrat et al. \(2024\)](#), [Idrees et al., \(2024\)](#) and [Roberts-Lombard and Petzer \(2024\)](#).

5. Discussion and conclusion

The digitization of financial markets, both globally and domestically, has led the banking industry toward a virtual banking model. Many researchers have contributed to the literature streams of the e-banking industry by applying different theories. Despite this, the literature revealed some gaps, particularly in the context of developing countries, such as the lack of a large sample size, customer segmentation based on Islamic and conventional banking and a perceived benefits analysis. This study addressed the gaps above to contribute to the literature on TAM as a broadband life cycle. Another uniqueness of this study is the inclusion of the variable of subjective norms.

Henceforward, the deviation between Intention and actual behavior is a natural phenomenon that exists due to the behavioral traits of any individual ([Fishbein and Ajzen, 1975](#)). These behavioral traits bridge the intention-behavior gap in decision-making and act as moderators and/or mediators. Therefore, to bridge the gap between Intention and Actual usage in the field of e-banking services, an augmented model of the TAM has been proposed in this study. Internationally, literature on technological acceptance theories has revealed a lack of consensus in measuring the customer's journey from intention to actual usage. To highlight the catalyst behind this journey, this study used a mediation-moderation analysis to conclude. Furthermore, a single has not been conducted addressing the intention-behavior relationship in a large sample size just like the studies of [Gu et al. \(2009\)](#); [Afshan et al. \(2018\)](#), [Arif et al. \(2016\)](#), [Ashrafi et al. \(2014\)](#), [Bhatti et al. \(2013\)](#), [Haider et al. \(2018\)](#), [Kazi and Mannan \(2013\)](#), [Rehman et al. \(2019\)](#), [Hussain et al. \(2020\)](#), [Toqeer et al. \(2021\)](#) and [Girish et al. \(2022\)](#).

A Web-based survey has been organized on the sample size of 2255 customers from all over Pakistan meeting the criteria ([Hair et al., 2019](#); [Comrey and Lee, 1992](#)). Adaptive measures from various research studies have been selected to inform the design of the questionnaire. Different statistical measures have been used to assess the contribution of variables across all models. Results revealed that CSR and OCR are marked as a significant negative resistance to the intention to adopt online banking services, likewise previous studies ([Kassim and Ramayah, 2015](#); [Afshan et al., 2018](#); [Akturan and Tezcan, 2012](#)). On the other hand, privacy risk has been labeled as insignificant resistance to influencing Intention to use online banking services following the trend of literature, likewise studies ([Chavali and Kumar, 2018](#); [Akturan and Tezcan, 2012](#); [Albashwahi and Motiwalla, 2019](#); [Hussain et al., 2020](#)). While FR and TR are explored as positive and significant factors in predicting intention to use, which shares dual association with the literature, such as perceived risks being significant, but the coefficient should be negative according to theories. Contrary to the literature, some previous studies, such as [Aboobucker and Bao \(2018\)](#) and [Foster et al. \(2022\)](#) found similar findings to this study. The insignificant impact of privacy risk on behavioral intention ($\beta = -0.001, p = 0.01$) aligned with studies signifying that privacy concerns may not generally matter in technology acceptance, mainly in contexts with strong established data protection laws ([Lee and Rha, 2016](#); [Phang et al., 2013](#)). Conversely, the null finding of this study aligns with the findings of [Ashrafi et al. \(2024\)](#) and [Aboobucker and Bao \(2018\)](#). Supporting the above statement, [Lowry et al. \(2017\)](#) concluded in a study that in collectivist cultures, users often prioritize practical utility over intellectual privacy risks ([Lowry et al., 2017](#)), as Pakistan is a collectivist-oriented society ([Hofstede, 2015](#)). The other

reason could be Pakistan's [Data Protection Act \(2023\)](#) which mandates data localization, user permission and breach forfeits may further attenuate privacy anxieties by fostering institutional trust but practically the situation is vice-versa, as studies of [Noorani \(2025\)](#) and [Ali and Hussain \(2024\)](#) highlighted data privacy act as a challenge in penetrating safest digital transactions. The other factor may be related to awareness or priorities regarding reading the "Data privacy policy," which is critically highlighted in the studies of [Amin and Khan \(2024\)](#). In this study, the majority of digital users reported that they do not read the privacy policies of digital platforms. This could explain why privacy risk did not significantly affect intentions to adopt, as users rely on legal frameworks rather than individual risk assessments. However, PEOU and PU have been identified as the most influential factors in adopting any technological advancement in previous studies. Parallel to risk, perceived benefits (PB) are another external variable. Internationally, numerous studies have been conducted, finding perceived benefits to be one of the significant reasons for using online services. According to the literature, the results are the same, as perceived benefits are labeled with a critical predictor. However, the negative coefficient of perceived benefits ($\beta = -0.107, p < 0.01$) is contrary to Technology Acceptance Theories, which may occur due to contextual differences specific to Pakistan. The same results were observed in a study by [Foster et al. \(2022\)](#), which found a negative coefficient for the perceived benefits of using Go-Pay services in the Malaysian context. Both countries share some demographic norms that may mask the associated risks, in contrast to the benefits, such as cyber-fraud, a low digital literacy rate and age-related factors.

Moreover, subjective norms have been found as a significant and positive factor influencing Intention to use online banking services, indicating consistency with the previous researches such as [Irimia-Diéguez et al. \(2023\)](#), [Pinochet et al. \(2019\)](#) in the Fintech sector and [Al-Okaily et al. \(2020\)](#) in digital payments vertical. In addition, the current study focused solely on the religious aspect of Islamic banking users, building on the studies of [Muslichah and Sanusi \(2020\)](#) and [Khan and Arif \(2022\)](#), which found that religious and social norms are positive and significant determinants of the Intention to use. Thus, the information shared about the social or religious influencers' group can drive potential customers toward using online banking.

Relying on the behavioral traits of customers, a bi-dimensional effect in terms of direct and indirect approaches has been used to test the hypothesis. The statistical results claim a direct and significant relationship between PRS with AU. Similarly, as a moderator, PRS is significant and bridges the gap between INT and AU. In mediation analysis, PL has been identified as an essential mediator between ITU and AU, indicating that PL can influence a customer's decision to use online banking services and, therefore, is considered a significant mediator in bridging the gap between ITU and AU. The results of current study indicated effect size as significant mediated effect of personality between intention-behavior, similarly the results of [Yang et al. \(2025\)](#). Based on the total effect of 0.394 in the students of China, it was found that the single mediation effect of action control accounted for 13.45%, in other words, the path of direct conversion of exercise intention into behavior through action control is the most significant, indicating that immediate execution is the core driving force of behavior transformation. Furthermore, [Chen et al. \(2022\)](#) found the same results with an effect size of 0.017 in the intention behavior, context of China. The findings may be similar in the Asian context, where users are forced to rely on personality-based self-regulation when embracing new technologies due to the lack of institutional trust brought about by the rapid digitalization of both markets. These similarities imply that personality mediates social risk management as well as individual behavior in transitional economies with strong

communal norms, which explains cross-cultural consistency even in the face of disparate political or economic systems.

Finally, determining the intention-retention cycle of the customer is one of the study's objectives. To measure the intention-retention journey of a customer, external variables such as satisfaction, service quality and continuous usage are considered. Satisfaction and service quality are included as independent variables, while continuous usage has been marked as the dependent variable. Furthermore, service quality has been added as a moderator between satisfaction and continuous usage as well. Results extracted that additive variables, such as service quality, moderate the relationship between satisfaction and continuous usage. The results of this study for service quality act as a moderator between satisfaction and continuous usage, which is significant with (0.071), which corroborates the prior studies for the same variables. Supporting the above statement, (Saengchai and Jermstittiparsert, 2020) conducted a survey of loyalty of customers in Thailand, where service quality acts as a moderator with values ($\beta = 0.237$, $p = 0.004$, t -values 2.920) also originating higher than the cutoff point. The same findings are quoted in the study by (Ashfaq *et al.*, 2020), where the chatbot agent's services moderate satisfaction, which indirectly affects continuous intention in the USA. These results suggest that, in the context of Asian markets, the moderating role of service quality may follow predictable patterns due to shared relational values rather than economic or technological differences alone. Furthermore, for direct effects, satisfaction and service quality are significant variables and quite enough to predict continuous usage in this study.

6. Recommendations and implications

The current study significantly extends the existing literature in the context of Pakistan regarding online banking services, as previous literature was flawed in exploring the continuance of adoption toward the latent advancement from a customer's behavioral perspective. Accordingly, this study endeavored to fill the void by examining exogenous variables and collaborating with TAM, termed as modified TAM. However, the statistical perspectives of this study support the basic inhibitors of TAM, as well as some external factors, such as subjective norms, PU, PEOU and perceived risk. As sequenced above, the study explored the perceived benefits as one of the significant variables in predicting the intention to adopt online banking services. Hence, the results suggest that, when it comes to swaying potential customers, marketers should also aim to raise customer awareness and highlight the benefits and efficacy of their services in the advertising messages they create for their target audience. This should be done in a way that highlights the ease of use and omnipresence of their product in the advertising message. Add-ons, marketing initiatives and educational programs highlighting the advantages of virtual banking over traditional banking should be launched to raise consumer awareness.

Consequently, e-commerce and fintech services are other technological innovations related to the financial market. The current research highlighted usefulness and ease of use as the basic significant inhibitors of online banking services. The proposed model can be applied to review fintech services from customers' and organizations' perspectives. According to a fintech network report (2021), a low literacy rate and lack of customer awareness are significant barriers to accessing fintech services. On that account, corresponding organizations can collaborate with academic institutions to initiate curriculum and internship programs that expedite the development of the FinTech industry, leading to a supplementary, comprehensive and scientifically innovative financial ecosystem in Pakistan.

Simultaneously, this study found a significant impact of subjective norms on the intention to adopt online services, advocating that managers should incorporate Islamic and social ideologies into their customer engagement and strategic planning practices. Augmenting

service offerings can be attained by consistent market exploration to identify varying customer preferences and cultural dynamics. Henceforth, cumulative trust and acceptance from latent customers can also be achieved by working with religious experts and community leaders. Broadly, the Islamic Fin-tech framework is another code-based technology in e-commerce, as Indonesia, Malaysia, Saudi Arabia and the UAE lead the global Fin-tech index (Global Islamic Fin-tech Report, 2021), while Pakistan ranks 10th in offering Islamic Fin-tech services. The same model of this study is directed toward the acceptance of the Islamic fintech framework from managerial and customer perspectives, aiming to expand the literature streams by ascertaining the impact of Islamic and social norms on the intention to adopt Islamic fintech services in a similar manner.

The most prominent result of the current study is the behavioral aspects of customers, as observed through pre-post analysis. The analyzed models in this study highlighted the antecedents of driving or constraining online banking services, as well as the process of moving from intention to use to continuous usage. Thus, by evaluating the pre-post behavioral phases of customers in conjunction with their retention state, AI-driven biometric solutions can be implemented by managers to enhance security and user-friendliness, mitigate perceived risks and foster client confidence. Aside from the above, AI can also be used to ensure rapid electronic fund transfers and withdrawals, attracting foreign patrons and enhancing remittance services. As mentioned above, the use of online banking services can be further enhanced by continuous observation and adaptation through artificial intelligence analytics. Furthermore, overseas customers often criticize the poor communication from bank personnel when they come to approach their local bank. For that purpose, customer relationship management (CRM) has substantial functions to satisfy customers through word-of-mouth. In Pakistan, poor CRM is a shortfall of the online banking industry, mainly due to a low literacy rate (Shafi *et al.*, 2013). To forge a convincing CRM, managers should generate a compelling content strategy and courtesy.

Based on current research, it is recommended that Financial regulatory bodies introduce contextual, culturally relevant and targeted strategies to promote online banking services in Pakistan. To maintain trust with potential customers, banks can integrate in-app transparency features that track transactions in real-time through security alerts. Trust campaigns can play a vital role in Pakistan, where people are influenced by their community and inspired by religious figures; Shariah compliance prompts can be a driving factor in building trust. Gender disparity in financial literacy rate is one of the demographic features of Pakistan and has been captured by the current study as well. Financial literacy programs that collaborate with female microfinance institutions and use AI-powered Urdu-language chatbots can enhance service quality and the inclusion of tailored financial literacy programs.

As, absence of rural areas in the sample profile is one of the limitations of this study, therefore, for rural inclusion in the potential customers, mobile bank vans with QR-code activation along with voice-guided banking chatbots can be provided to prevail awareness to adopt online banking services, because the mostly population of rural areas is even not able to read Urdu language.

Some researchers have identified the need for further research in the area of behavioral-technological approaches addressing behavioral traits for e-services. Personality and planning have been introduced as external variables to bridge the intention-behavior gap. So, this study constructed an updated and the latest version of TAM with the collaboration of the Theory of Intention-Behavior gap named augmented TAM. By analyzing the proposed model, the authors commend anticipating market changes and accommodating the diverse risk tolerance and inclinations of online banking customers, using sentiment analysis and dynamic risk management strategies. Furthermore, legal bodies can tailor financial services and learning

supplies to sanction clients and uphold market stability, retaining a close eye on market data and customer behavior to iteratively improve prediction models for improved online banking risk management as well. As mentioned above, the same model can be applied to evaluate the investment trends of cryptocurrency investors. Some investors have a state-oriented personality, while others have an action-oriented personality. Therefore, the planning of investors may affect the buying and selling trend of cryptocurrencies. Consequently, managerial staff can predict the buying or selling trend of investors in response to any event, thereby stabilizing the market. The same criteria apply to stock market analysis.

6.1 Limitations and further studies

Pursuing staging foremost constituents in the intention to adopt online banking services, the study encountered innumerable barriers and limitations calling for further studies. Primarily, this study included a large sample of the population, despite the limited representation of rural areas due to restricted access and the low literacy rate of the population. Next in order, people were hesitant in providing their data throughout Web surveys due to security concerns, which led to a lack of corroboration to minimize legitimacy concerns.

In particular, the technological barrier is a significant factor behind the low adoption rate of online services in Pakistan, particularly in rural areas, due to inadequate infrastructure and limited internet access. Therefore, the authors advocate for further studies on the impact of technological barriers on the adoption of online services in rural areas. In the bargain, a gender-discrimination analysis should be piloted, such as examining the propensity of men's acceptance of technological advancements compared to women, as well as the drivers and resistance behind it.

In Pakistan, due to security concerns, banks do not disclose information or statistics on the actual usage of any customer on their official websites. Thus, the authors call for further research by granting special consent from legal bodies to use data on the actual usage of e-banking services. Last but not least, a comparative study should be conducted in-depth on the intention-retention cycle of Islamic and conventional online banking customers to highlight disparities between these two cohorts.

While the current study provides detailed insights into the intention-retention phenomenon of online banking services in the Pakistani context, several limitations have been identified throughout the research, which should be addressed in future studies. In addition, a cross-sectional study should be used in future research to measure variation in behavioral and other factors across different periods among the same respondents, thereby precluding causal inferences about the relationship between variables.

Gender imbalances in the sample profile indicate another limitation, which warrants further research to identify the distinct access barriers faced by women. To bypass self-reported bias, the same study should be conducted by a mixed-methods design, such as interviews and Web surveys, prioritizing gender-balanced respondents.

Finally, the construct of subjective norms in the current study does not fully capture the behavioral aspects and religious and social influences in defining the range of intention to adopt online banking services. This feature can be counted as an agenda for future research.

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Authors' contributions

Idea for the article: Mahpara Naeem developed the idea to write a research paper.

Literature search: Dr Syed Tehseen Jawaid searched and synthesized the literature review to

extract related research papers for the study. *Data analysis*: Mahpara Naeem and Dr Syed Tehseen Jawaid analyzed and synthesized data to conduct statistical tests. *Drafting*: Mahpara Naeem drafted the work. *Critically revision*: Dr Sadaf Mustafa critically revised the work.

Note

- [1.] (Rahi *et al.*, 2022; Smith, 2019; Salem *et al.*, 2019; Oertzen, 2019; Farah *et al.*, 2019; Siyal *et al.*, 2018; Yuan *et al.*, 2018; Zhang *et al.*, 2018; Ayo *et al.*, 2016; Gan *et al.*, 2017; Garcia *et al.*, 2020; [Glavee-Geo et al., 2017](#); Kassim *et al.*, 2015) considered the same criteria to determine discriminant validity in the studies of the same context.

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